THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** sonority	Sonorous
2	**(b)** gold	1 g gold → 2 km wire
3	**(d)** gallium	Low melting point
4	**(b)** Incorrect — iodine is a lustrous non-metal	Exception
5	**(d)** diamond	Strong 3-D network
6	**(a)** 4	Two pairs
7	**(c)** catenation and tetravalency	Two key properties
8	**(c)** C_6H_6	Cyclic
9	**(c)** Incorrect — it does not ionise	Covalent compound
10	**(a)** naphthalene	High carbon content

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They lose electrons easily to form positive ions.	2
12	(i) gold, silver (ii) gold, copper	1+1
13	They are covalent and do not form ions.	2
14	Diamond — cutting/jewellery; graphite — lubricant/electrodes	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $2\,Zn + O_2 \longrightarrow 2\,ZnO$ (ii) $3\,MnO_2 + 4\,Al \longrightarrow 3\,Mn + 2\,Al_2O_3$ + heat (iii) $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$	1+1+1
16	Saturated: single bonds only, blue flame, substitution reactions; unsaturated: double/triple bonds, sooty flame, addition reactions	3
17	Connect a metal wire/strip in a circuit with a cell and bulb; the bulb glows, showing current passes through the metal.	3
	OR: Strong electrostatic forces between ions need much energy to overcome; NaCl (1074 K), $MgCl_2$ (981 K)	1+2
18	Ethyne C_2H_2, propyne C_3H_4, butyne C_4H_6	1+1+1
	OR: Soap forms insoluble scum with Ca^{2+}/Mg^{2+}; more soap needed; use detergents or soften water	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) removal of gangue from ore (ii) roasting $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$; reduction $ZnO + C \longrightarrow Zn + CO$ (iii) carbon is a cheap reducing agent	1+3+1
	OR: (i) A (ii) prevents air dissolving in water (iii) absorbs moisture (iv) both air and water are needed for rusting (v) painting/galvanising	1+1+1+1+1
20	(i) chloro/bromo prefix; –ol, –al, –one suffixes replacing final 'e' of alkane (ii) bromoethane; propanal	3+2
	OR: (i) esterification (ii) $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (iii) catalyst/dehydrating agent (iv) saponification — gives ethanol and sodium ethanoate (v) perfumes/flavouring	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** magnesium hydroxide and hydrogen	$Mg + 2\,H_2O \longrightarrow Mg(OH)_2 + H_2$
2	**(b)** Student A is correct	ZnO reacts with both
3	**(d)** hydrogen	$Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$
4	**(d)** gangue	Removed during enrichment
5	**(c)** potassium	Most reactive
6	**(a)** C_nH_{2n-2}	Triple bond
7	**(b)** Student A is correct	CH_2 = 14 u
8	**(d)** ethanoic acid	5–8%
9	**(b)** sunlight	Photochemical
10	**(b)** propanol	–ol

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Insoluble impurities that settle at the bottom below the anode during electrolytic refining.	2
12	False — aluminium is used; its protective oxide layer prevents further reaction, and it is a good conductor of heat.	2
13	It is dehydrated to ethene: $C_2H_5OH \longrightarrow CH_2{=}CH_2 + H_2O$	2
14	False — ethane is an alkane (C_nH_{2n+2}); ethene is an alkene (C_nH_{2n}).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metals form basic oxides and lose electrons; non-metals form acidic/neutral oxides and gain electrons; metals displace hydrogen from dilute acids, non-metals do not.	3
16	A: ethanoic acid; $CH_3COOH + C_2H_5OH \xrightarrow{\text{acid}} CH_3COOC_2H_5 + H_2O$; ethyl ethanoate (ester)	1+1+1
17	Enrichment (concentration) of ore → conversion to oxide (roasting/calcination) → reduction to metal → refining	3
	OR: To protect it from rusting; even if scratched, zinc being more reactive corrodes first, protecting iron.	1+2
18	(i) CH_3COOH (ii) CH_3COCH_3 (iii) $CH_3CH_2CH_2Cl$	1+1+1
	OR: C^{4-}: nucleus with 6 protons cannot hold 10 electrons; C^{4+}: removing 4 electrons requires very high energy. Hence carbon shares electrons.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Na 2,8,1; Cl 2,8,7 (ii) $Na \longrightarrow Na^+ + e^-$; $Cl + e^- \longrightarrow Cl^-$; $Na^+\,Cl^-$ (iii) high melting point; conducts in molten state/solution	2+2+1
	OR: (i) (a) $4\,Na + O_2 \longrightarrow 2\,Na_2O$ (b) $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (c) $Fe + 2\,HCl \longrightarrow FeCl_2 + H_2$ (ii) gold/silver	4+1
20	(i) self-linking to form chains (ii) C–C bonds are strong and stable (iii) hexane, 2-methylpentane, cyclohexane	1+1+3
	OR: (i) $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$; $C_2H_5OH \xrightarrow{\text{conc}\cdot H_2SO_4} C_2H_4 + H_2O$ (ii) affects nervous system, liver (iii) blended with petrol (gasohol); clean fuel	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** aluminium	$Al_2O_3 \cdot 2H_2O$
2	**(a)** Correct	Cu is more reactive than Ag
3	**(d)** chromium and nickel	Resists rusting
4	**(c)** 2	2, 8, 2
5	**(b)** Student A is correct	Pure Cu deposits on cathode
6	**(a)** each C is bonded to four others in a 3-D network	Rigid structure
7	**(a)** Correct	Animal fats contain saturated fatty acids
8	**(c)** ethyne	$HC{\equiv}CH$
9	**(b)** CO_2	Brisk effervescence
10	**(b)** Student A is correct	Tail is hydrophobic

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$	2
12	Lead, copper, silver, gold (any two)	1+1
13	Six C atoms in a ring by single bonds, each C bonded to 2 H: 6 C–C + 12 C–H = 18 covalent bonds.	2
14	Oxygen is added (and hydrogen removed) from ethanol.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) ionic (ii) XY_2 i.e. $MgCl_2$ (iii) yes	1+1+1
16	(i) decolourises bromine water (ii) effervescence with $NaHCO_3$ (iii) Na gives H_2 bubbles	1+1+1
17	Basic copper carbonate (moist CO_2); Ag_2S (sulphur compounds); cleaned with acidic substances like lemon/tamarind	1+1+1
	OR: Salt + hydrogen: $Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2$; hydrogen — burns with a pop sound	1+1+1
18	CO_2, H_2O, heat and light; they release large energy; LPG/CNG	1+1+1
	OR: Speeds up the reaction without being consumed; nickel/palladium; vanaspati ghee (saturated fat)	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $Fe_2O_3 \cdot x\,H_2O$ (ii) bronze is an alloy resistant to corrosion (iii) homogeneous mixture of metal with other elements (iv) copper and tin (v) increases hardness/corrosion resistance	1+1+1+1+1
	OR: (i) oxides reacting with both acids and bases (ii) $ZnO + 2\,HCl \longrightarrow ZnCl_2 + H_2O$; $ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (iii) CO or H_2O / N_2O	1+3+1
20	(i) incomplete combustion deposits carbon (ii) air holes are blocked (iii) butane (iv) produces poisonous CO (v) $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$	1+1+1+1+1
	OR: (i) (a) ethanoic acid turns blue litmus red; ethanol no change (b) acid gives CO_2; alcohol no reaction (c) ethanol pleasant; acid vinegar smell (ii) CH_3CH_2OH; CH_3COOH	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** oxygen	Used in respiration
2	**(b)** Ag_2S	Silver sulphide
3	**(c)** electrolytic reduction	Electrolysis of molten chlorides
4	**(d)** in molten state and aqueous solution	Free ions
5	**(b)** basic copper carbonate	Moist CO_2
6	**(a)** 3	CH_3COCH_3
7	**(b)** C_6H_{12}	Saturated ring
8	**(c)** ethene	Conc. H_2SO_4, 443 K
9	**(a)** long-chain carboxylic acids	Fatty acids
10	**(b)** 2	n- and iso-butane

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ions are held in fixed positions in the crystal and cannot move.	2
12	A flux combines with gangue to form a fusible slag that can be removed.	2
13	CO_2; turns lime water milky	1+1
14	Compound of carbon and hydrogen only; ethane; ethene	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Electrolysis of molten Al_2O_3: cathode $Al^{3+} + 3\,e^- \longrightarrow Al$; anode $2\,O^{2-} \longrightarrow O_2 + 4\,e^-$	1+1+1
16	(i) CH_3OH (ii) HCHO (iii) CH_3COCH_3	1+1+1
17	Metals listed in decreasing order of reactivity; Na > Zn > Fe > Cu > Ag	1+2
	OR: (i) sonorous (ii) good conductor, ductile (iii) non-metal that conducts electricity	1+1+1
18	Graphite: hexagonal layers slide over each other; diamond: rigid 3-D network, hardest substance	1½+1½
	OR: Soap forms micelles (large clusters) giving a colloid; ionic heads on the surface repel each other, so micelles stay apart	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) impure Cu anode, pure Cu cathode, acidified $CuSO_4$ electrolyte, anode mud below (ii) anode: $Cu \longrightarrow Cu^{2+} + 2\,e^-$; cathode: $Cu^{2+} + 2\,e^- \longrightarrow Cu$	3+2
	OR: (i) lustre, malleability, ductility, conductivity (ii) mercury is liquid; sodium/potassium are soft; lead is a poor conductor (iii) utensils/wires/machinery	2+2+1
20	(i) H_2: single bond; O_2: double bond; N_2: triple bond — electron sharing (ii) weak intermolecular forces	3+2
	OR: (i) soaps: Na/K salts of long-chain carboxylic acids; detergents: ammonium/sulphonate salts of long-chain acids (ii) hydrophobic tails inward in oil, ionic heads outward (iii) their charged ends do not form insoluble precipitates with Ca/Mg ions	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.